

Digital Material Passport

ID **DMP-METAL-003** - Version **1.0.0**
Issue Date **2025-05-15** - Certificate Type **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	Precision Aerospace Inc. Aviation Boulevard 789 Mountain View 94043 US materials@precision-aero.example.com	Order ID PO-34567 - Pos. 3 - 2025-04-25 - Quantity 500 kg Delivery ID DN-89012 - Pos. 1 - 2025-05-14 - Quantity 500 kg

Product

Product Name	Aluminum Alloy 7075-T6	Name (AA)	7075-T6
Batch ID	H-43210-01	Number (UNS)	A97075
Surface Condition	Rolled	Shape	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm
Weight	500 kg	Product Norm	AMS 4045 (2023)
Production Date	2025-05-12		
Country of Origin	DE		

Chemical Analysis

Heat Number **H-43210** - Melting Process **VAR** - Casting Method **VacuumCasting** - Casting Date **2025-05-10** - Sample Location **Ladle**

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
Al	%			89.7	Cu	%	1.2	2	1.5
Zn	%	5.1	6.1	5.6	Cr	%	0.18	0.28	0.22
Mg	%	2.1	2.9	2.4	F1		0.18	0.28	0.22

F1 = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.22

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength¹	Rm	570 / 572 / 574 MPa - Mean 572 - Min 570 - Max 574 - Std Dev 2	530 MPa		ASTM E8	✓
0.2% Yield Strength¹	Rp0.2	503 / 505 / 507 MPa - Mean 505 - Min 503 - Max 507 - Std Dev 2	480 MPa		ASTM E8	✓
Elongation¹	A	10.8 / 11.2 / 11 % - Mean 11 - Min 10.8 - Max 11.2	10 %		ASTM E8	✓

¹ 3 specimens tested

Physical Properties

Property	Symbol	Actual	Target/Min	Maximum	Method	Status
Density	ρ	2.81 g/cm³	2.81 g/cm³		ASTM B311	✓
Coefficient of Thermal Expansion Temperature range: 20-100°C	α	23.4 10 ⁻⁶ /K	23.5 10 ⁻⁶ /K		ASTM E228	✓
Thermal Conductivity At 25°C	λ	130 W/(m·K)	120 W/(m·K)		ASTM E1461	✓
Specific Heat Capacity At 25°C	cp	862 J/(kg·K)	860 J/(kg·K)		ASTM E1269	✓
Electrical Resistivity At 20°C	ρe	0.0538 μΩ·m		0.055 μΩ·m	ASTM B193	✓
Poisson's Ratio	ν	0.33	0.33		ASTM E132	✓
Melting Range	Tm	477 - 635 °C	475 - 635 °C		ASTM E1142	✓
Relative Magnetic Permeability	μr	1.00002		1.0001	ASTM A342	✓
Surface Roughness	Ra	0.8 μm		1.6 μm	ISO 4287	✓
Emissivity At 25°C	ε	0.09		0.11	ASTM E408	✓
Surface Tension	γ	0.875 N/m	0.87 N/m		ASTM D971	✓
Diffusion Coefficient	D	2.3E-9 m²/s	2.2E-9 m²/s		ASTM E1559	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with AMS 4045 and the specified test methods. All results are within the specified limits.

Validated by **Elsa Müller**, Materials Engineer, Quality Assurance - 2025-05-15